

DreamHandoff: Stabilizing Reactive Action-Chunk Selection and Asynchronous Bank Transitions

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Abstract

In the DREAM-Chunk setting, asynchronous reactive action-chunk execution creates three distinct temporal problems: within-bank selection persistence, activation-time grounding of replacement dreams, and action continuity across bank transitions. DreamHandoff adapts DREAM-Chunk’s candidate sampling, world-model imagination, and observation-conditioned latent matching using SmolVLA and R2Dreamer on an SO-101 robot. Candidate matching provides limited ranking information and often weak separation between the nearest two candidates. Calibrated absolute hysteresis makes the stability/reactivity trade-off explicit: on saved prospective trajectories, selector replay reduces within-bank switches from 17.44% to 7.45%, with a measurable hold-out recorded-continuation proxy cost. We show structurally that forecasting reactive selection without future observations changes observation-versus-prediction into prediction-versus-prediction. Separately, DreamHandoff grounds delayed suffixes at the observed activation posterior rather than the stored request-time latent. For bank transitions, offline paired reconstruction shows that coherent same-index outgoing prefixes recover 84.1% of the command-seam benefit and 90.3% of the action-difference-seam benefit of hindsight factual prefixes under matched delay.

1 Introduction

Action-chunk policies produce a sequence of commands from one observation, allowing expensive inference to support faster robot control. This interface is useful for vision-language-action models (VLAs), but observations can change while a chunk executes. A controller must reconcile the temporal coherence of a sampled plan with the need to respond to the robot’s actual trajectory.

DREAM-Chunk [2] addresses this tension by sampling several candidate chunks, imagining each candidate’s latent trajectory, and selecting the action whose phase-aligned dreamed state best matches the current observation. Its critical distinction is between an *observation-conditioned live state* and *predicted candidate states*. New observations provide an external reality anchor: they can make a different candidate more compatible with the realized trajectory. Asynchronous policy inference is already part of DREAM-Chunk.

In this DREAM-Chunk setting, we separate two observed temporal phenomena: candidate-index instability within an active bank and action-command discontinuity at asynchronous bank replacement. Candidate ranking contains limited information, but runner-up separations are often small, and prospective replay shows frequent switches and $A \rightarrow B \rightarrow A$ reversals. How can selection remain reactive without switching on every small ordering change?

Separately, canonical action-space finite differences are substantially larger at bank boundaries than at strict interiors (figure 1), motivating explicit action-continuity management at replacement.

At a replacement request, DREAM-Chunk stores the request-time latent state to initialize the incoming bank’s dreams. During inference, the current bank continues executing, and real observations and actions update the live posterior. DreamHandoff instead crops elapsed candidate actions and starts the surviving dreams from that newer posterior when the replacement bank activates. This aligns their latent starting point with the time at which their actions can execute.

A separate information problem arises at request time: the outgoing bank’s future reactive action path is still unknown, because future observations and selector decisions have not yet occurred. Recursively forecasting that selector would substitute predictions for those observations and can make the followed candidate self-consistent with its own dream. We ask whether one coherent outgoing candidate can guide the incoming policy without predicting the exact future switch sequence.

We therefore distinguish three responsibilities. *Within-bank selection persistence* uses hysteresis to manage the stability/reactivity trade-off. *Activation-time latent grounding* initializes incoming dreams from the posterior conditioned on intervening observations and actions; it does not itself enforce action continuity. *Between-bank action continuity* uses coherent prefix conditioning to guide the incoming policy before the exact future reactive path is known.

DreamHandoff implements this separation with SmolVLA, an adapted R2Dreamer, and an SO-101 robot. Its contributions are:

- an analysis of candidate coverage and decisiveness, distinguishing nearest distance from runner-up separation and quantifying limited ranking signal and weak separation;
- calibrated hysteresis and an empirical accounting of its within-bank stability/reactivity trade-off;
- an activation-time re-grounding adaptation and, separately, an analysis of model-only future-selector forecasting and open-loop request-to-handoff prediction;
- a characterization of bank-boundary command discontinuity and a paired prefix-conditioning evaluation in which Static recovers most of the matched-delay factual-prefix benefit.

The contribution is an execution adaptation and empirical study, not a new candidate-selection paradigm or a task-success benchmark. The frozen evidence combines physical capture, selector replay, and offline policy reconstruction; their roles are kept distinct throughout.

2 Background and related work

Action chunks and VLAs. Action Chunking with Transformers learns distributions over action sequences for manipulation [9]. SmolVLA combines visual and language conditioning with an action-generating model and describes an asynchronous inference stack [7]. We use one SmolVLA checkpoint fine-tuned through upstream LeRobot [3].

Reactive latent matching. DREAM-Chunk supplies the core bank-execution mechanism: sampling multiple chunks, imagining their futures, and choosing phase-aligned actions by latent matching [2]. Its asynchronous algorithm stores a latent state when requesting the replacement bank and uses that stored state to initialize the new dreams. DreamHandoff retains the sampling and matching pattern but changes activation grounding and studies repeated selection and bank seams. The present evidence does not reproduce or extend DREAM-Chunk’s task-success comparisons.

Latent world models. R2-Dreamer learns a recurrent world model using redundancy reduction without an observation decoder [6]. DreamHandoff adapts its image/vector encoders, block-GRU recurrent state-space model, categorical posterior, dynamics prior, and feature construction. The focused trainer combines dynamics KL, representation KL, and a Barlow Twins objective [8]; it includes no decoder or control objective. This model architecture and representation-learning machinery are inherited work, not DreamHandoff contributions.

Continuity during latency. Real-Time Chunking (RTC) conditions generation on an outgoing action prefix so that a new flow-policy chunk can be produced during execution [1]. We use the RTC processor in the pinned LeRobot implementation [4]. The question here is specific to reactive banks: which prefix is useful when future observation-driven switches are unknown? Same-index conditioning retains one coherent outgoing candidate for each incoming candidate. The evaluation compares that choice with hindsight factual prefixes and a deployment-style schedule, rather than claiming a new RTC algorithm.

3 Method

3.1 Reactive candidate-bank execution

Let o_t denote the current observation and $u_{t-1} \in \mathbb{R}^6$ the previously served canonical action. The live recurrent posterior is updated by

$$s_t = \mathcal{O}(s_{t-1}, u_{t-1}, o_t), \quad (1)$$

where $\mathcal{O}$ includes the observation-conditioned R2 update. At an episode reset the recurrent state is cleared and the previous action is masked on the first observation. A *candidate* is one policy-generated action chunk; a *bank* contains $N = 10$ candidates from a shared prepared observation, sampled using independent SmolVLA flow noise in one batch. Each original chunk has $K = 50$ actions. Policy-space actions are retained for prefix guidance, while canonical six-dimensional actions are used for dispatch and world-model imagination.

Write the usable bank activated at action count b as $A_i[p]$, where $i \in \{1, \dots, N\}$ and $p = t - b$ is its phase. Candidate dreams begin from the activation posterior:

$$\tilde{s}_{i,0} = s_b, \quad \tilde{s}_{i,p+1} = F(\tilde{s}_{i,p}, A_i[p]). \quad (2)$$

Here F includes canonical-action normalization and deterministic R2 imagination using categorical modes. Dreaming branches a copy of the posterior and leaves the live recurrent state unchanged. Each dreamed state in the selector is the state *before* its corresponding action, avoiding a one-step phase offset.

For stochastic state z and deterministic state h , the matching feature is

$$\phi(s) = [\text{vec}(z); h] \in \mathbb{R}^{2560}, \quad 2560 = 32 \cdot 16 + 2048. \quad (3)$$

The distance and memoryless nearest index are

$$D_i(t) = \|\phi(s_t) - \phi(\tilde{s}_{i,p})\|_2, \quad n_t = \arg \min_i D_i(t). \quad (4)$$

The selected command is $u_t = A_{j_t}[p]$. For $p > 0$, memoryless selection sets $j_t = n_t$; equal minima use the lowest candidate index. At phase zero all dreams share the same posterior, so the implementation instead draws a uniform initial index from a rollout-wide CPU Torch generator seeded once with zero. Episode resets do not rewind that stream. The incumbent is bank-local and resets at activation.

3.2 Candidate geometry and hysteresis

Sort the distances independently at each row. The nearest distance $d_1(t)$ characterizes the nearest candidate’s absolute match to the live posterior in the learned representation. The runner-up separation $d_2(t) - d_1(t)$ quantifies the nearest candidate’s distance advantage. The two quantities answer different questions: proximity to a candidate is not decisiveness of the argmin. Neither is a calibrated physical or task error.

For incumbent $c = j_{t-1}$ in the same bank, absolute hysteresis sets

$$j_t = \begin{cases} n_t, & D_c(t) - D_{n_t}(t) > \tau, \\ c, & \text{otherwise,} \end{cases} \quad (5)$$

with the strict inequality and released threshold

$$\tau = 0.06851652264595032. \quad (6)$$

The incumbent advantage in equation (5) is not generally the runner-up gap: the incumbent can have any current rank. Hysteresis explicitly exchanges immediate nearest-distance optimality for persistence. It neither interpolates actions nor modifies the candidate trajectories. A change of index within one active bank is a *switch*; replacement by a new bank is a *handoff*, not a switch between indices of the previous bank.

3.3 Activation-time re-grounding

The first bank is generated synchronously. Later, a request saves the prepared observation o_r and served-action count r while the active bank continues executing. There is at most one replacement request pending or in flight. When the result

activates at count b , its actual elapsed delay is $\ell = b - r$. If $B_i[0 : K]$ is the returned request-time chunk, the usable suffix is

$$H = \min(35, K - \ell), \quad A_i[p] = B_i[\ell + p], \quad 0 \leq p < H. \quad (7)$$

Thus the controller discards one leading action per action served during inference and caps the remainder at 35 actions. It then initializes [equation \(2\)](#) using the *actual live posterior* s_b , which has incorporated intervening real observations and executed actions. Phase restarts at zero. Empty suffixes and expiration of an active bank before replacement raise errors.

The information flow is consequently

Request at r	Save o_r ; generate $B_{1:N}$ asynchronously
During $[r, b)$	Execute, observe, and update the live posterior
Activation at b	Crop by $b - r$; imagine suffixes from s_b

DREAM-Chunk initializes a replacement bank’s dreamed trajectories from the latent state stored when the policy request was issued; it does not forecast an activation posterior for this purpose. DreamHandoff instead removes actions that elapsed during inference and imagines the surviving candidate suffixes from the live posterior available when the bank activates. That posterior incorporates intervening real observations and executed actions and is temporally aligned with execution of those suffixes. The action proposals still depend on o_r , and re-grounding does not enforce action continuity. We treat this as an architectural adaptation; this study contains no controlled task-level ablation establishing universal superiority over request-time grounding.

3.4 Why forecasting the future selector is different

When the incoming policy is requested at r , its action-level prefix could in principle follow the outgoing bank’s future reactive path during $[r, b)$. Predicting that path requires forecasting selector decisions based on future observations, executed actions, and live posteriors that do not yet exist at r . Replacing the unavailable observation-conditioned posterior with a model rollout changes the selector’s inputs:

$$\begin{aligned} \text{online: } s_t^{\text{obs}} &\leftrightarrow \tilde{s}_{i,p}, \\ \text{forecast: } \hat{s}_t &\leftrightarrow \tilde{s}_{i,p}. \end{aligned}$$

Here s_t^{obs} is conditioned on future real observations, whereas $\hat{s}_t$ and the phase-aligned candidate state $\tilde{s}_{i,p}$ in the forecast are both predicted by the world model. Consider a deterministic idealization in which a forecast follows candidate i :

$$\hat{s}_{t+1} = F(\hat{s}_t, a_i[t]), \quad \tilde{s}_{i,t+1} = F(\tilde{s}_{i,t}, a_i[t]). \quad (8)$$

If the two full latent states coincide, identical dynamics and actions preserve coincidence. Nearby states may remain close, depending on the dynamics. The followed candidate can therefore remain self-consistent with the forecast used to select it. The real-observation residual that could redirect selection is absent.

This is a structural self-consistency or candidate-locking tendency, not a theorem that arbitrary stochastic rollouts lock. Equality need not hold; numerical effects, model dynamics, and alternative rollout procedures can matter. Ordinary

DREAM-Chunk remains reactive because real observations update its live posterior. The concern arises specifically when forecasting that future observation-conditioned selector before those observations exist. The open-loop handoff-prediction diagnostic in [section 5.4](#) provides complementary empirical context; it does not test candidate locking.

3.5 Prefix-conditioned handoffs

For prefix guidance at request time, the factual future outgoing path is unavailable. Recursively forecasting the future selector would require the model-on-model substitution described above. The open-loop endpoint analysis in [section 5.4](#) provides complementary evidence that alternative request-to-handoff propagations are only weakly separated in the learned representation. Rather than depending on an exact predicted reactive switch sequence, DreamHandoff tests whether one coherent outgoing candidate can provide sufficient prefix guidance. Prefixes are provided in policy space to LeRobot’s RTC processor. DreamHandoff uses its linear attention schedule and maximum guidance weight 10; these are inherited generation mechanics.

Plain generates from the request-time observation without outgoing-prefix information. **GT** supplies the factual actions later executed during the delay, repeated across incoming candidates. It asks how much seam improvement prefix conditioning could provide if the actual future outgoing path were known in hindsight; it is privileged, nondeployable information, not an optimal-control oracle. **Static** supplies each incoming candidate with the same-index outgoing candidate’s coherent action prefix. It asks whether a path already available at request time can recover much of the factual-prefix benefit without forecasting the future reactive selector sequence. For Static and GT, prefix length, RTC inference delay, and execution horizon equal the realized delay L_e : matched-delay Static controls scheduling while varying prefix information, using retrospectively known delay.

Static-live separately reconstructs deployment-style timing and taper semantics: a 15-action outgoing prefix starting at the request phase, the recorded predicted inference delay, and execution horizon 15. The predictor converts the maximum observed policy latency to control ticks by taking the ceiling; a warmup generation supplies the initial estimate. For the linear RTC schedule, prefix weights are one before the predicted delay (clamped to the execution horizon), taper between that delay and the horizon, and vanish thereafter. When delay and horizon coincide, there is no intervening taper. The initial bank remains unguided. Actual elapsed delay still determines the activation crop in [equation \(7\)](#).

Reactive is a separate SmolVLA reference inferred from the activation-time observation for diagnostics. It is not an optimal-control oracle. Physical collection used Plain generation; all prefix-conditioned conditions, including Static-live, are offline reconstructions of the saved handoffs. Their comparison isolates local generated-action seams under recorded contexts rather than evaluating the closed-loop trajectories those controllers would induce.

4 Experimental setup

4.1 System, data, and evidence

The task is to pick up an orange rectangle, align its two holes with two pegs, and insert it using one SO-101 setup. The study uses one SmolVLA policy checkpoint and one R2Dreamer checkpoint. There are 120 demonstration episodes with a fixed 96/24 training/model-validation split. SmolVLA was fine-tuned on the 96 training episodes through upstream LeRobot. The 24 validation episodes also provide the offline candidate-proxy matrix, split into eight selector-calibration and sixteen held-out evaluation episodes; they are distinct from the subsequently collected prospective physical episodes.

R2 receives context and wrist images, each resized to 64×64 and concatenated into six channels, plus the unscaled six-dimensional robot state. Canonical actions are normalized using checkpoint q01/q99 values before entering its dynamics. The adapted subset comes from NM512/r2dreamer at revision 546e4fab, under the MIT license [5]. Preprocessing and runtime integration are DreamHandoff adaptations; the architecture remains attributed to R2Dreamer.

The saved prospective capture contains 24 physical episodes, 12,257 control rows, 454 activated banks, and 430 non-initial handoffs. Collection used the frozen hysteresis threshold, activation-time re-grounding, and Plain asynchronous generation. The configuration targets 30 control ticks per second, with refill requested when at most 15 usable actions remain. These settings do not establish a measured end-to-end control frequency. The collection protocol targets saved successful episodes, so this evidence is not a success-rate sample or a comparison of task success across controllers.

Three evidence types must be distinguished. Offline calibration uses generated candidate banks at fixed origins in recorded demonstrations. Prospective selector replay changes selector decisions while holding the captured posterior/distance sequence fixed; it does not simulate the changed physical observations that alternative actions would produce. Prefix reconstruction resamples incoming banks on the same saved handoffs using paired generation noise. Its Plain replay gate reproduces the captured bank suffixes exactly in both policy and canonical action space for all 430 events. This verifies reconstruction fidelity for Plain, not physical outcomes for guided conditions.

4.2 Metrics and calibration protocol

Recorded-continuation proxy. At offline bank origin b and phase p , let x_{b+p+1}^{rec} be the next recorded robot state. Candidate proxy error and selected regret are

$$e_i(b, p) = \left\| A_i[p] - x_{b+p+1}^{\text{rec}} \right\|_2, \quad (9)$$

$$R(b, p) = e_{j_t}(b, p) - \min_i e_i(b, p). \quad (10)$$

Top-1 agreement asks whether nearest-by-R2 selection agrees with $\arg \min_i e_i$. The minimizer is an oracle only for the factual recorded continuation. These records contain no counterfactual next states for unexecuted candidates.

Threshold selection. Calibration generates ten 50-action chunks at origins `range(0, episode_length, 35)` and evaluates a legal 35-phase bank horizon. Policy noise comes from one rollout-wide CUDA Torch generator seeded zero, in sorted episode/origin order; phase-zero selection uses the CPU stream described above. On calibration rows with $p > 0$, incumbent advantages from a $\tau = 0$ replay define 45 thresholds: 40 linearly spaced points from zero through the advantage q90, plus q95, q97.5, q99, q99.5, and q99.9. Points are Pareto-filtered by simultaneous minimization of switch rate and mean proxy regret. Among Pareto points whose mean regret is no more than 5% above memoryless selection, the rule chooses the lowest switch rate. The threshold is frozen before prospective collection.

Switching and uncertainty. Calibration switch rate is the mean switch indicator across the split’s rows, with bank-initial rows contributing zero. Prospective operational rate instead counts switches over consecutive within-bank transitions after excluding the phase-0-to-1 transition. This exclusion respects the special phase-zero initialization. A rapid successive switch is within three control rows of the previous switch in that bank; an $A \rightarrow B \rightarrow A$ reversal occurs on consecutive control rows. The prospective matching-distance penalty is the operational-row mean of $D_{j_t} - d_1$, distinct from the recorded-continuation regret in equation (10).

Held-out uncertainty uses a paired bootstrap over 228 (episode, bank origin) units, 2,000 resamples, seed zero, and the 2.5/97.5 percentiles. Handoff endpoint and paired seam intervals instead use 5,000 physical-episode-clustered resamples with seed zero. These uncertainty summaries describe the recorded sample under their respective clustering assumptions; they do not account for model retraining or variation across tasks.

Command-space seams. For a handoff at b , let u_{b-1}, u_{b-2} be factual outgoing commands and v_i the first incoming command after cropping candidate i . The reported event-level seams are

$$C_b = \frac{1}{N} \sum_i \|v_i - u_{b-1}\|_2, \quad (11)$$

$$Q_b = \frac{1}{N} \sum_i \|(v_i - u_{b-1}) - (u_{b-1} - u_{b-2})\|_2. \quad (12)$$

Results average these quantities over the paired handoffs. Thus they average over incoming candidates, rather than taking the minimum or only the factual phase-zero draw. C_b is the command seam and Q_b the action-difference seam. Both are canonical command-space quantities, not physical velocity, acceleration, jerk, or measured smoothness.

Reproducibility. Code, fixed configurations, canonical JSON outputs, figure sources, exact external artifact revisions/hashes, and reproduction commands are available in the [DreamHandoff repository](#) (DreamHandoff v1.0.0). All displayed scientific figures are unchanged release PDFs. The manuscript build requires only files in this paper repository; it does not run the experiments or load external checkpoints.

Candidate-bank behavior and chunk handoffs

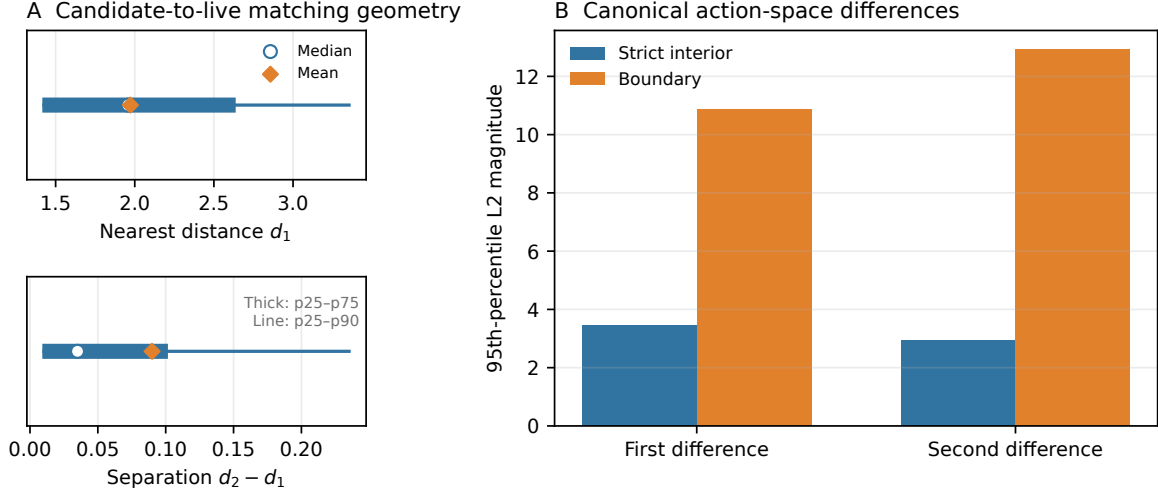


Figure 1: Two empirical temporal problems in asynchronous reactive chunking. A: nearest distance d_1 reflects candidate coverage of the live posterior; runner-up separation $d_2 - d_1$ characterizes decisiveness over 11,803 rows ($p > 0$). Thick/thin segments span p25–p75/p25–p90; circles/diamonds mark medians/means. Weak separation and replay switches motivate examining within-bank selection persistence. B: elevated p95 canonical action-space finite differences at bank boundaries versus strict interiors (phases 10–39 of the 50-action representation) motivate between-bank action continuity. These are neither calibrated task errors nor physical motion measurements.

5 Results

5.1 Candidate ranking is informative but weakly separated

Nearest-by-R2 selection agrees with the best of ten candidates under the recorded-continuation proxy on 18.255% of 11,854 valid offline rows: 20.0% on calibration and 17.4% on held-out rows. The nominal uniform-choice rate is 10%. This contains measurable but limited candidate-selection information. It is not evidence for a strong selector, a task-success predictor, or reliable counterfactual candidate ranking.

On 11,803 operational physical-capture rows ($p > 0$), nearest distance d_1 characterizes the nearest candidate’s absolute match to the live posterior, while runner-up separation $d_2 - d_1$ characterizes the decisiveness of the argmin winner (figure 1A). Runner-up separations are small relative to absolute nearest distances: mean separation is 0.090 versus mean nearest distance 1.97, and median separation is 0.035 versus median nearest distance 1.96. Their p90 values are 0.236 and 3.36, respectively. This scale comparison is descriptive, not a per-row normalized margin. Small gaps alone do not prove switching; together with the direct switch and reversal measurements below, they motivate hysteresis as a stability/reactivity trade-off.

5.2 Hysteresis exposes a stability/reactivity trade-off

On calibration episodes, the selected threshold changes switch rate from 23.61% to 13.51% and mean recorded-continuation regret from 0.855 to 0.895 (figure 2, left). Held-out switch rate changes from 23.72% to 12.89%, with observed regret increase +0.0502. The paired bank-clustered bootstrap mean regret increase is 0.0501, with 95% interval [0.0297, 0.0719]. The observed delta and bootstrap mean are different statistics.

Table 1: Prospective selector replay. Operational switch rates use 11,350 within-bank transitions.

Metric	Memoryless	Hysteresis
Operational switches	1,980	846
Operational switch rate	17.44%	7.45%
Rapid successive switches	1,050	257
$A \rightarrow B \rightarrow A$ reversals	163	33
Mean matching penalty	0	0.0050

The positive interval supports a proxy cost in this sample; the calibration constraint does not guarantee the same relative regret bound holds on held-out data.

Prospective replay gives a complementary result (table 1 and figure 2, right). Across 11,350 operational transitions, memoryless selection makes 1,980 switches (17.44%), versus 846 (7.45%) for hysteresis. Rapid successive switches fall from 1,050 to 257 and consecutive reversals from 163 to 33. The mean prospective latent matching-distance penalty is 0.0050. This penalty quantifies departure from the current nearest candidate; it does not contradict the larger held-out proxy cost, because the metrics and evidence populations differ.

The result demonstrates persistence under fixed recorded observations. It does not establish that the discarded switches would have been harmful to physical execution, or that hysteresis would improve task success. The mechanism deliberately delays small-advantage switches, including potentially useful corrections. That trade-off is visible rather than hidden behind a generic claim of smoothing.

5.3 Bank boundaries have larger command discontinuities

The physical capture exhibits a separate transition problem (figure 1B). First-difference magnitudes $\|u_t - u_{t-1}\|_2$ have boundary/interior means 4.13/1.39 and p95 values 10.86/3.45.

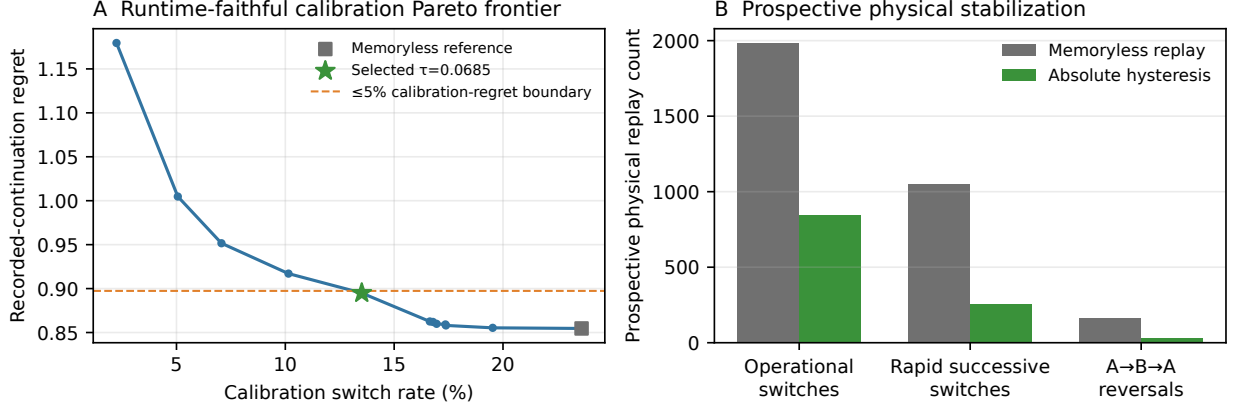


Figure 2: Within-bank stability has a measurable cost. Left: calibration Pareto frontier and the selected absolute threshold, using recorded-continuation proxy regret and the 5% calibration-regret constraint. Right: selector replay on saved prospective physical trajectories. Operational switches exclude phase-0-to-1 transitions; rapid switches occur within three rows of a preceding switch, and reversals are consecutive $A \rightarrow B \rightarrow A$ patterns. The two panels use different evidence and switch-rate denominators. The memoryless replay is not a separately collected robot condition.

Second-difference magnitudes $\|u_t - 2u_{t-1} + u_{t-2}\|_2$ have means 4.65/1.36 and p95 values 12.93/2.94.

The strict interior mask uses phases 10–39 of the 50-action representation, a nominal 10-action margin at either end of that representation. It is applied to the available recorded phases; it does not redefine the runtime’s 35-action cap or impose a 10-action buffer before an early replacement. All commands involved in an interior finite difference must satisfy the mask and belong to one bank. Episode boundaries are excluded. A second-difference stencil is classified as boundary when it crosses a bank origin, producing 859 boundary stencils rather than one value per handoff. The first-difference boundary sample contains 430 values.

These descriptive contrasts motivate treating bank replacement explicitly. They do not isolate a causal effect of handoff timing, and they differ from the reconstructed seam averages in equation (12): the characterization uses factual dispatched commands, while reconstruction averages over all incoming candidates. This accounts for the distinct Plain command-seam summary (4.10) and factual first-difference boundary mean (4.13).

5.4 Open-loop handoff prediction

The handoff-prediction diagnostic asks how strongly and reliably open-loop propagation of alternative action paths across $[r, b)$ distinguishes the posterior eventually observed at handoff. For each of 430 non-initial handoffs, the target is s_b , which incorporates real observations. The *exact factual history* comparator rolls the model from s_r over the canonical actions actually executed in $[r, b)$; those actions are privileged hindsight information. Static comparators instead roll each of the ten outgoing candidates’ request-to-activation action paths from the same request posterior.

Two selection rules yield different static comparators. *Target-selected best-of-ten* chooses the static endpoint nearest the actual activation posterior, retrospectively. *Action-path-nearest* chooses the static path closest to the factual executed sequence before evaluating endpoint error. If $v_{j,k}$ is static candidate j ’s canonical command and u_k the factual com-

mand over the interval, its path discrepancy is

$$m_j = \sqrt{\frac{1}{\ell} \sum_{k=0}^{\ell-1} \sum_{a=1}^6 (v_{j,k,a} - u_{k,a})^2}. \quad (13)$$

Factual path mismatch is $\min_j m_j$; the action-path comparator uses $\arg \min_j m_j$. Endpoint error is L2 distance in the R2 matching representation. The signed effect is *static endpoint error minus exact-history endpoint error*, so positive favors exact-history propagation.

Mean factual path mismatch increases from zero with no switches to 0.764 with one and 1.157 with at least two (table 2 and figure 3). Yet closer reproduction of executed actions does not ensure the smallest endpoint error. Overall mean errors are 1.77 for target-selected static, 1.81 for action-path-nearest static, and 1.80 for exact history. The overall signed differences are -0.0221 for target selection and $+0.0178$ for action-path selection, with the intervals in table 2.

The three overall endpoint means span only 0.04 in the same R2 matching space. For scale, operational nearest-to-live distances have mean 1.97 and median 1.96, while runner-up separations have median 0.035 and mean 0.090 (figure 1A). These quantities come from different evaluation contexts and are an intuitive scale comparison, not a formal normalized statistic; the alternative open-loop propagation paths differ only weakly in this latent representation.

For zero-switch events, the factual action history itself is one static candidate, so hindsight target selection is structurally no worse than that candidate, apart from numerical effects. Its negative difference is not evidence that factual actions harm prediction. In switch cases, action-path-nearest static has higher mean endpoint error than exact history, while target-selected static retains a negative mean difference with intervals spanning zero. Comparator choice therefore changes the apparent winner.

Across these comparisons, exact-history and static-candidate propagation exchange ordering with the static selection rule, and the absolute differences are small. Even privileged exact-history propagation is not uniformly closest to the eventual observed posterior. Open-loop world-model

Table 2: Open-loop request-to-handoff endpoint prediction by number of selector switches. Effects are mean static-minus-exact-history endpoint errors, with 95% episode-clustered bootstrap intervals (5,000 resamples). The zero-switch action-path effect rounds to zero; the JSON retains floating-point residuals.

Switches	Events	Mean path mismatch	Target-selected effect [95% CI]	Action-path-nearest effect [95% CI]
0	255	0	−0.0243 [−0.0367, −0.0142]	0 [0, 0]
1	119	0.764	−0.0151 [−0.0367, 0.0070]	0.0380 [0.0154, 0.0651]
≥ 2	56	1.157	−0.0268 [−0.0566, 0.0046]	0.0558 [0.0237, 0.0885]
All	430	0.362	−0.0221 [−0.0342, −0.0110]	0.0178 [0.0090, 0.0285]

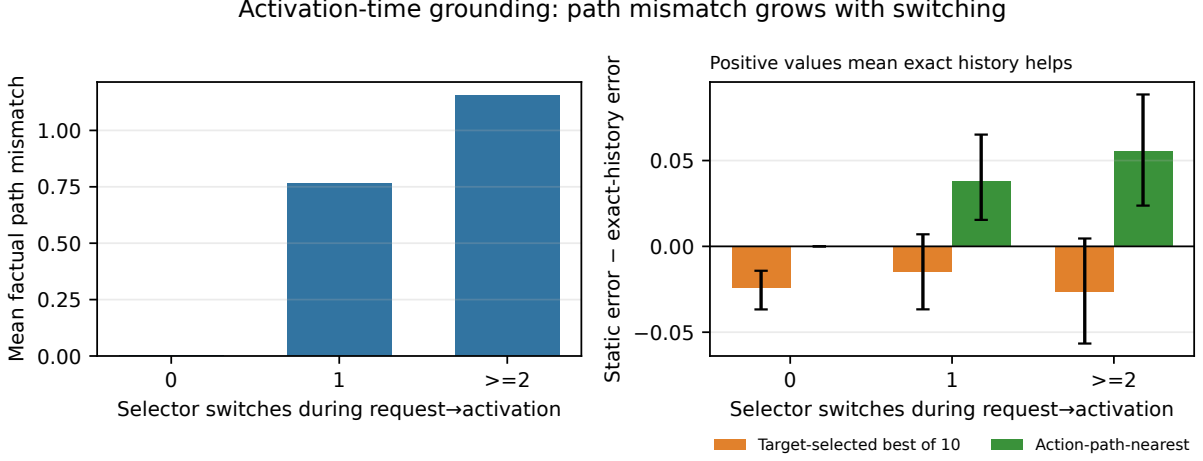


Figure 3: Open-loop handoff-prediction diagnostic. Factual action paths depart farther from any single static candidate as request-to-activation switches accumulate (left). Endpoint comparisons change ordering with the static selection rule (right; 95% episode-clustered intervals). Target selection uses the actual activation posterior in hindsight; action-path selection uses factual action similarity. The diagnostic concerns future-path prediction for prefix guidance, not bank grounding; the embedded plot title is unchanged from the frozen figure.

propagation therefore does not provide a uniquely reliable or strongly discriminative surrogate for the eventual observed handoff posterior. Separately, the increasing action-path mismatch with selector switches shows why a single static candidate need not reproduce the factual outgoing trajectory. The diagnostic does not test candidate locking or compare bank-grounding choices. The following experiment asks a different question: whether reproducing that factual path is necessary at all for obtaining most of the handoff-continuity benefit.

5.5 Coherent prefixes recover most hindsight seam benefit

Across all 430 paired handoffs, mean command/action-difference seams are 4.10/4.79 for Plain, 2.77/3.31 for Static, and 2.52/3.15 for GT (figure 4 and table 3). The Static-minus-Plain command-seam effect is −1.333 with 95% interval [−1.478, −1.198]; its action-difference effect is −1.478 with interval [−1.628, −1.338]. These paired episode-clustered summaries support a seam reduction under the fixed reconstruction protocol.

To quantify how much factual-prefix benefit is recovered, for either seam metric M compute

$$\rho_M = \frac{\overline{M}_{\text{Plain}} - \overline{M}_{\text{Static}}}{\overline{M}_{\text{Plain}} - \overline{M}_{\text{GT}}}. \quad (14)$$

Under matched delay, Static captures 84.1% of GT’s command-seam benefit and 90.3% of its action-difference-

Table 3: Mean reconstructed seams. Static and GT share the realized-delay schedule; Static-live uses the recorded deployment-style schedule.

Condition	Command	Action difference
Plain	4.10	4.79
Static	2.77	3.31
GT (hindsight)	2.52	3.15
Static-live	2.10	2.29

seam benefit relative to Plain. Computed from full-precision means, these are ratios of aggregate improvements, not averages of per-event ratios. Much of the handoff-continuity benefit available from knowing the factual future outgoing path can therefore be recovered with one coherent candidate path, without reconstructing the exact future observation-driven switch sequence in this setting. GT still has lower mean seams; Static need not reproduce the factual path or recover every benefit of factual information.

The Static-live deployment-style reconstruction gives means 2.10 and 2.29 versus Plain’s 4.10 and 4.79. It uses recorded delay predictions and the actual RTC taper semantics, rather than retrospective equality of prefix length, delay, and horizon. Its smaller raw values than GT cannot establish that it is causally better, because the schedules differ. Moreover, its runtime availability as an option does not turn this offline reconstruction into a separately tested physical controller.

Mean candidate diversity spans approximately 0.823–

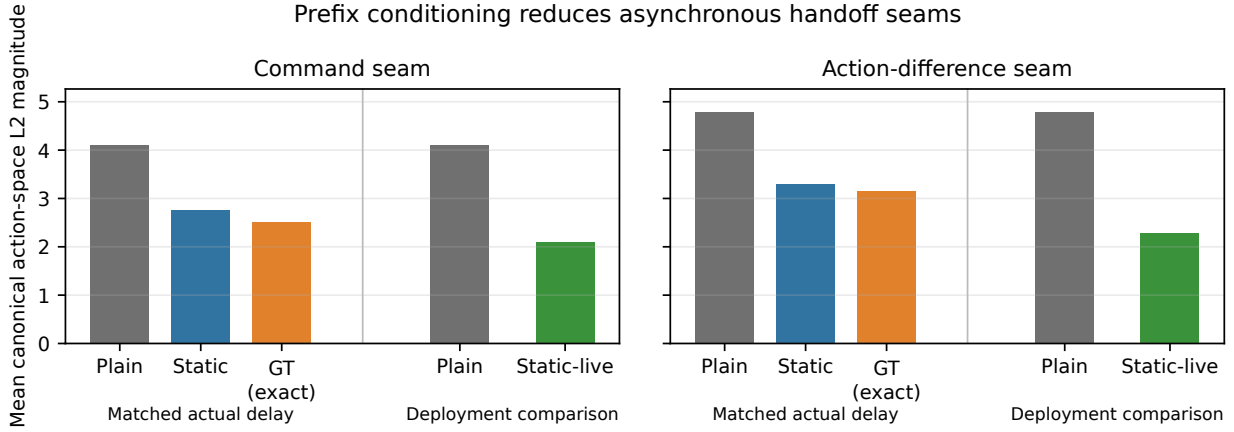


Figure 4: Prefix-conditioned bank transitions. Means over 430 paired offline handoffs, with incoming-candidate averaging within each event. In each panel, the left group compares Plain, Static, and hindsight factual-prefix GT under matched actual delay; the right compares Plain with the deployment-style Static-live schedule. The repeated Plain bars are the same baseline. Static-live uses a different prefix length and taper schedule from GT, so their raw means do not support a causal ranking. All quantities are in canonical action space.

0.877 across these four conditions. Here diversity is mean unordered-pair root-mean-square canonical action distance over the surviving generated trajectories (averaging over time and action coordinates), not the nearest-to-live latent metric. This diagnostic gives no indication of a large collapse in aggregate action diversity in the reported conditions. It does not establish preserved task-relevant mode coverage. The state-seam diagnostic averages the L2 distance between each candidate’s first incoming command and the observed robot state at activation. Static-live’s state-seam diagnostic also does not justify a generic state-continuity claim. The principal supported result remains the paired command-space seam reduction.

6 Discussion and limitations

We distinguish within-bank selection persistence, activation-time latent grounding, and between-bank action continuity. Nearest latent matching supplies some candidate-ranking information, yet many runner-up separations are small. Hysteresis makes the cost of retaining an incumbent explicit, reducing repeated switches and reversals while accepting increased recorded-continuation proxy regret. It is not a claim that all switches are undesirable. A large correction and a weak-margin index change can have different roles, and the present proxy cannot determine their physical value.

DREAM-Chunk initializes replacement-bank dreams from the latent stored when the policy request is issued. DreamHandoff instead crops actions that elapsed during inference and grounds the surviving suffixes in the live posterior available when the bank activates. The motivation is temporal alignment: that posterior incorporates the real observations and actions that occurred while policy inference was running. We treat this as an architectural adaptation rather than a demonstrated universal improvement over request-time grounding.

A different issue arises when deciding what outgoing trajectory should guide the incoming policy at request time. The factual future reactive path is unknown because it depends on future observations, live posteriors, and selector decisions. Recursively imagined states would produce a model-on-model selector that can become self-consistent

with the candidate it follows; this structural tendency does not imply universal locking or a flaw in ordinary observation-anchored DREAM-Chunk. The endpoint diagnostic provides complementary context: even privileged factual-action propagation and static-candidate propagation differ only slightly and exchange ordering with the comparator (section 5.4). This weak discrimination provides additional empirical context for avoiding dependence on an exact predicted switch sequence for prefix guidance.

Plain provides no outgoing-prefix information; GT quantifies the seam benefit available if the factual outgoing path were known in hindsight, without serving as an optimal-control oracle. Matched-delay Static uses one coherent same-index candidate prefix and recovers most of GT’s benefit in this setup. This is the empirical answer to the future-path information problem for local command seams: reconstructing the exact observation-driven switch sequence is unnecessary for much of the observed reduction. This interpretation depends on paired schedule control (figure 4). Static-live reconstructs deployment-style scheduling and taper semantics for comparison with Plain; their difference from GT’s prevents direct causal ranking.

Several limitations constrain the scientific scope. First, there is one task, one SO-101 robot setup/family, one policy checkpoint, and one world-model checkpoint. Effects can depend on the policy’s sampled alternatives, the learned representation, the task, latency distribution, chunk length, and robot dynamics. No cross-task or cross-checkpoint generalization is established. The finite-sample bootstrap intervals do not measure this broader variation.

Second, no task-success comparison was collected between controller variants. Prospective memoryless replay reuses posteriors produced under the factual hysteresis controller; alternative commands would generally change later observations. Prefix-conditioned conditions likewise reuse factual contexts and outgoing histories. They measure local reconstructed seams, not closed-loop outcomes under repeated conditioned replacements. The saved-episode collection protocol further limits any attempt to infer an unconditional success rate.

Third, the recorded-continuation proxy ranks commands

against the next factual recorded state. It cannot reveal what would have happened had another candidate been executed. Representation-space matching distances also depend on the R2 feature construction and training; their numerical scale has no direct task-error interpretation. Hysteresis’s calibrated threshold should therefore not be assumed portable across models.

Finally, command-space finite differences and seams do not establish physical smoothness or jerk reduction. Static-live’s state-seam result limits even a general claim about state continuity, and its scheduling mismatch with GT prevents direct causal ranking. There is no frozen ablation establishing universal superiority of activation-time over request-time grounding, nor an empirical test proving universal candidate locking. A subsequent physical study should compare controller variants under a common outcome protocol and measure task completion and robot motion directly, while varying representation and latency separately.

7 Conclusion

DreamHandoff separates three temporal responsibilities in the DREAM-Chunk setting. Hysteresis stabilizes within-bank candidate selection while real observations are available. Activation-time re-grounding uses the newer observation-conditioned posterior that actually exists when a replacement bank becomes active, after cropping actions elapsed during inference. Prefix conditioning addresses a different information problem: the factual future reactive path is unavailable when the new policy is requested, so a coherent outgoing candidate guides generation without recursively forecasting the unknown observation-driven switch sequence. In this setup, that Static prefix recovers most of the matched-delay command-seam and action-difference-seam benefit available from hindsight GT conditioning. These findings concern one task and one robot/policy/world-model configuration; we demonstrate no task-success improvement, physical smoothness, or universal superiority of activation-time over request-time grounding.

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